

Let's Flip the Switch!



Hamid Sutton, Terry Townsend, Brad Kivett



Target Areas for Redesign

- Physical Appearance
- Software Features
 - News
 - Theme Settings
 - Shop
- New Implementations
 - Friends list
 - Messaging
 - Custom Themes
 - Power Settings

Issues with Target Areas

- Physical Attributes
 - Screen is too small
 - Buttons are small
 - Controllers are hard to hold
 - System doesn't fit well inside the Dock
 - Hidden SD card
 - Kickstand quality

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- Software features

- News and Shop
 - Search Bar
 - Organization
- Theme Settings
 - Limited Theme Customization
- Gallery
- Power Settings
 - Only "Sleep Mode"

Issues with Physical Attributes



Tight fit may
cause scratching

Flat design on
controller

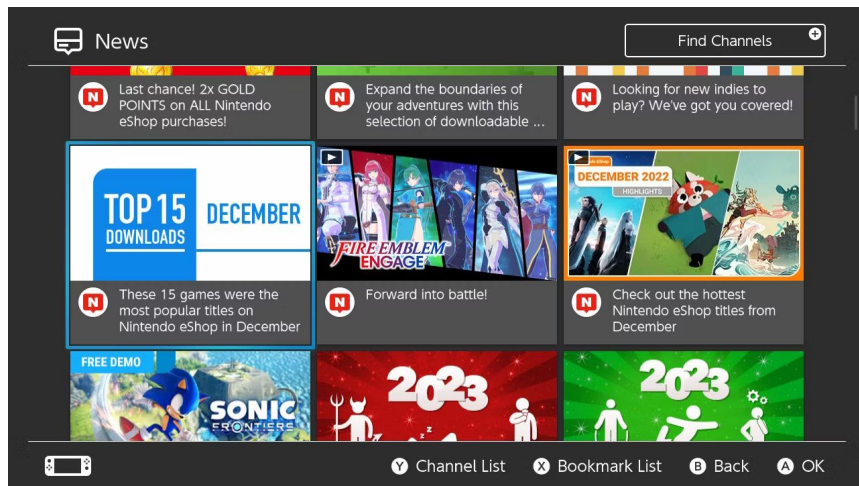


Hidden SD
Card Slot

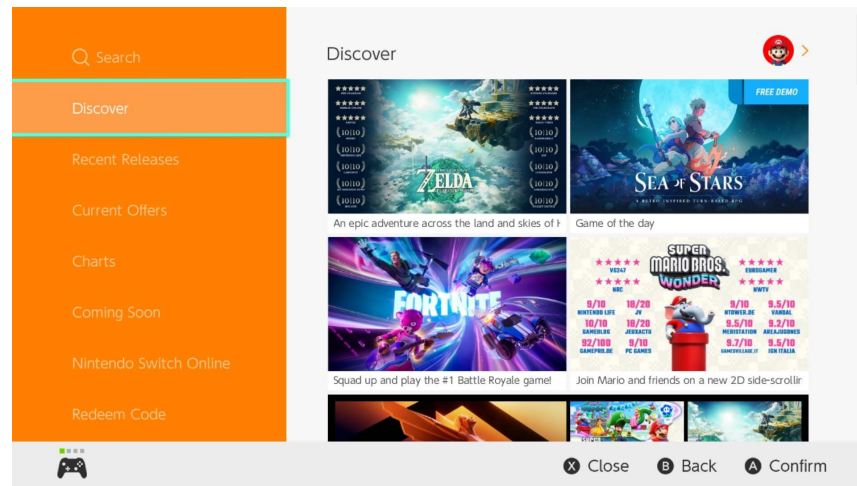


Issues with Software Features

News Page



eShop Page





Contextual Inquiry and Analysis

Representative Tasks

- Turn on the console
- Attach the remotes to the console
- Change the theme of the device
- Find “Pokemon Unite News”
- Purchase “Super Mario Bros 3”
- Play a game for 1-3 minutes
- Prop the device like this (demonstrated)
- Locate the memory card slot
- Mirror the Switch to the TV
- Power off the controllers

Do's and Don'ts



Do's

- Make the joy-con and buttons larger
- Add lighting to controls
- Redesign the kickstand
- Increase the size of switch
- Improve the look of themes
- Redesign the docking station
- Change SD Slot location

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Don'ts

- Change the game card size
- Difficult to navigate
- Make it difficult to open the kickstand
- Non-user friendly design

Artifact Model



Artifact Model (Continued)

User had some trouble locating the kickstand.
(P3: Line 136; P5: Line 115; P7: Lines 54-56; P9: Lines 54-56)

User spent too much time locating memory card (hidden)
(P3: Line 134, P7: Lines 54-56; P9: Lines 53-58)

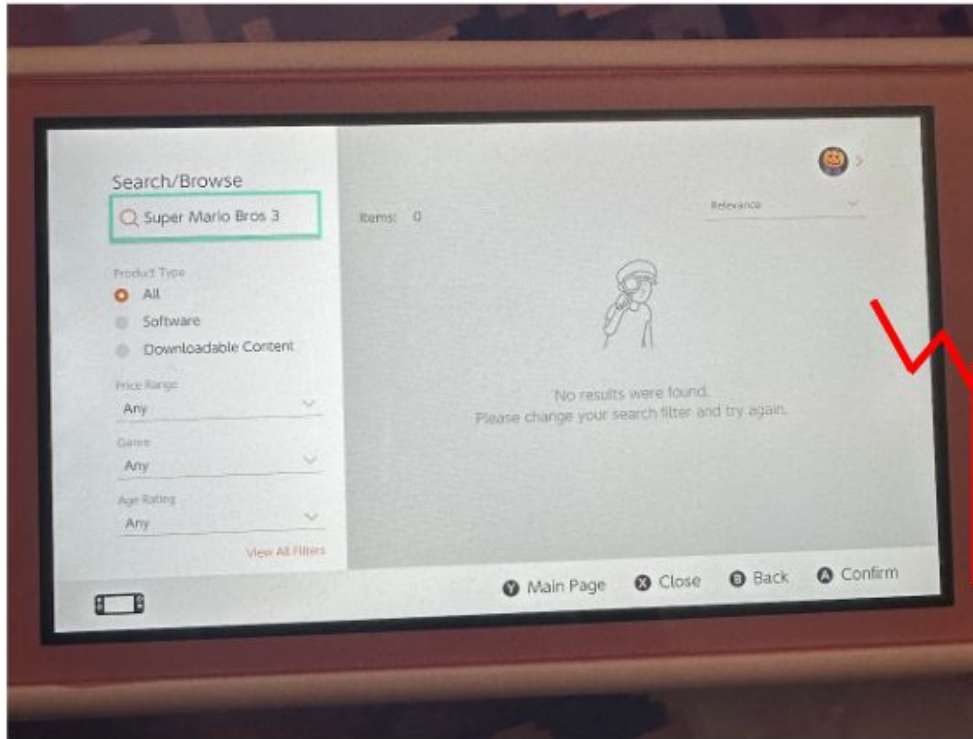


Artifact Model (Continued)



Users had some trouble connecting switch to the dock. (P1: Line 113, P3: Lines 86-89).

Artifact Model (Continued)

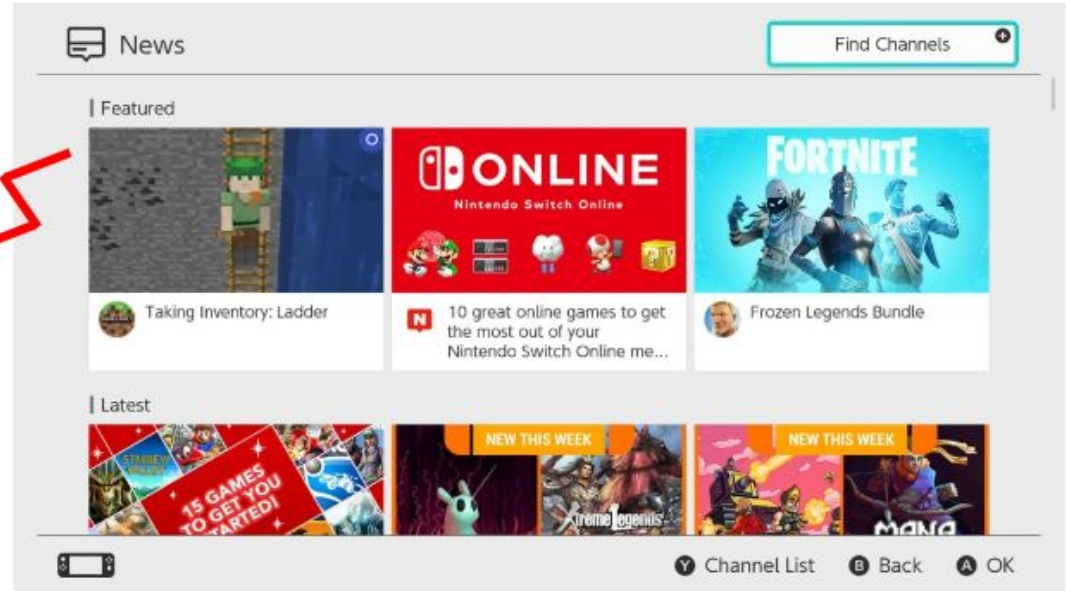


Users could not successfully find "Super Mario Bros. 3" on the shop when it is downloadable via the NES game pack. (P7: Lines 45-47, P8: Lines 40-42, P9: Lines 43-46)

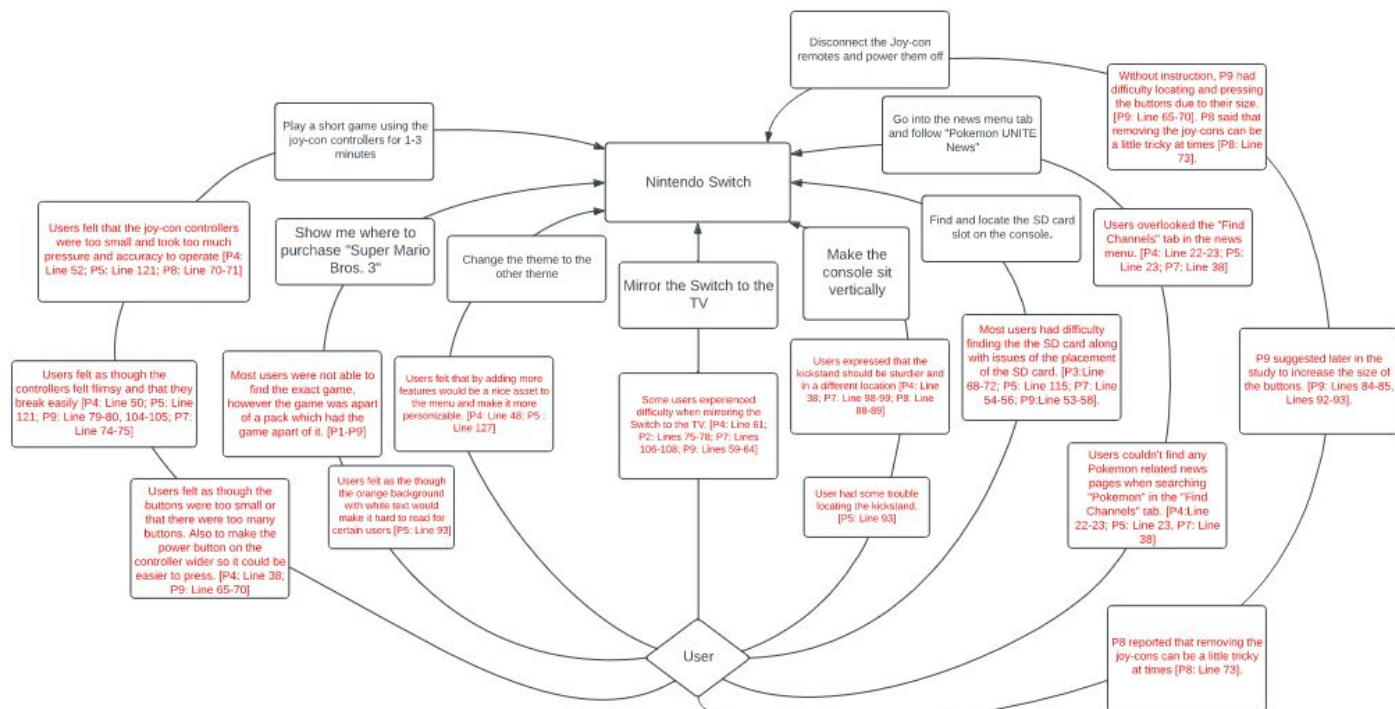
Artifact Model (Continued)

Users experienced difficulty finding “Pokémon UNITE” when searching “Pokémon” with an “e” instead of the “é” [P4: Lines 22-23, P5: Line 23, P7: Line 38, P8: Lines 36-39; P9: Line 41-42].

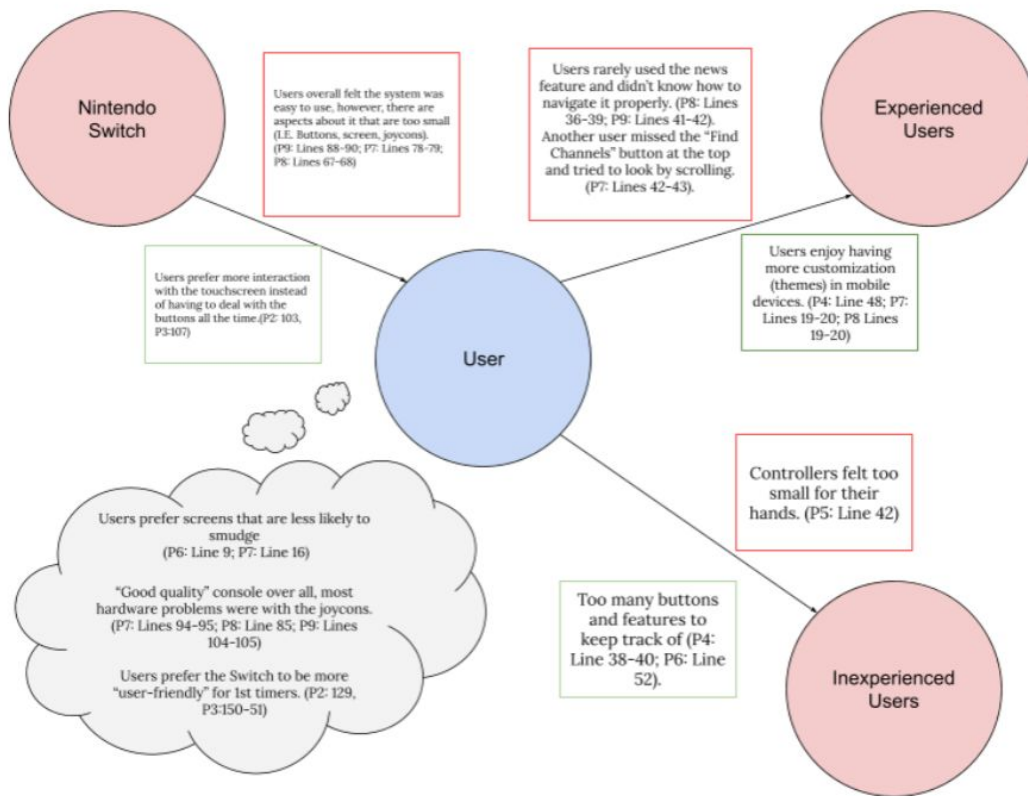
Another user missed the “Find Channels” button at the top and tried to look by scrolling. [P7: Lines 39-43]



Flow Model



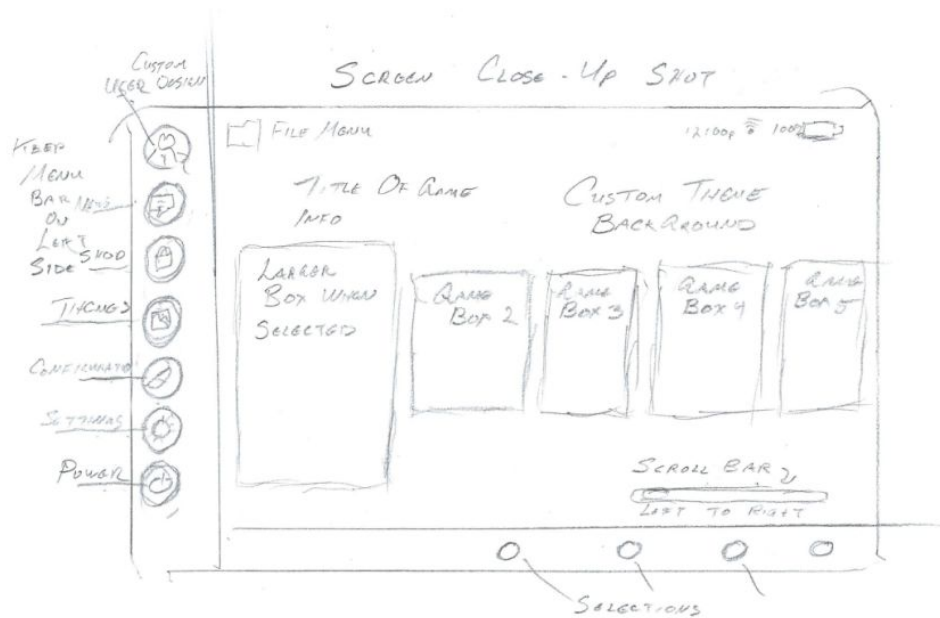
Social Model



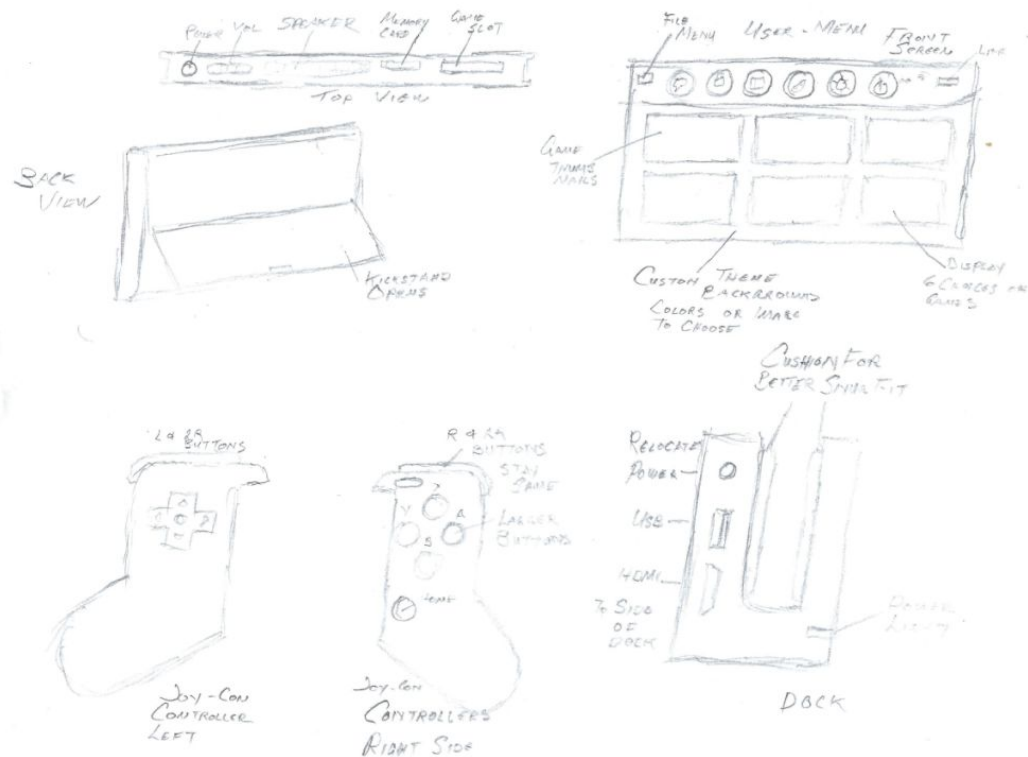


Sketches & Final Paper Prototype

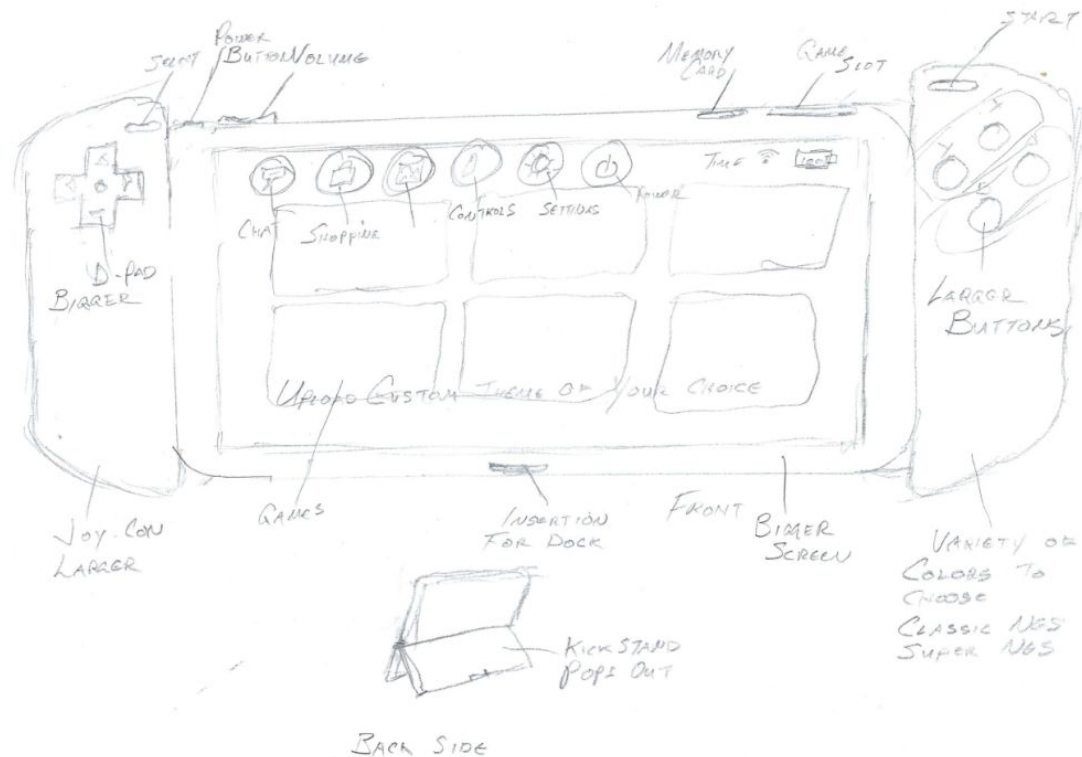
Sketches



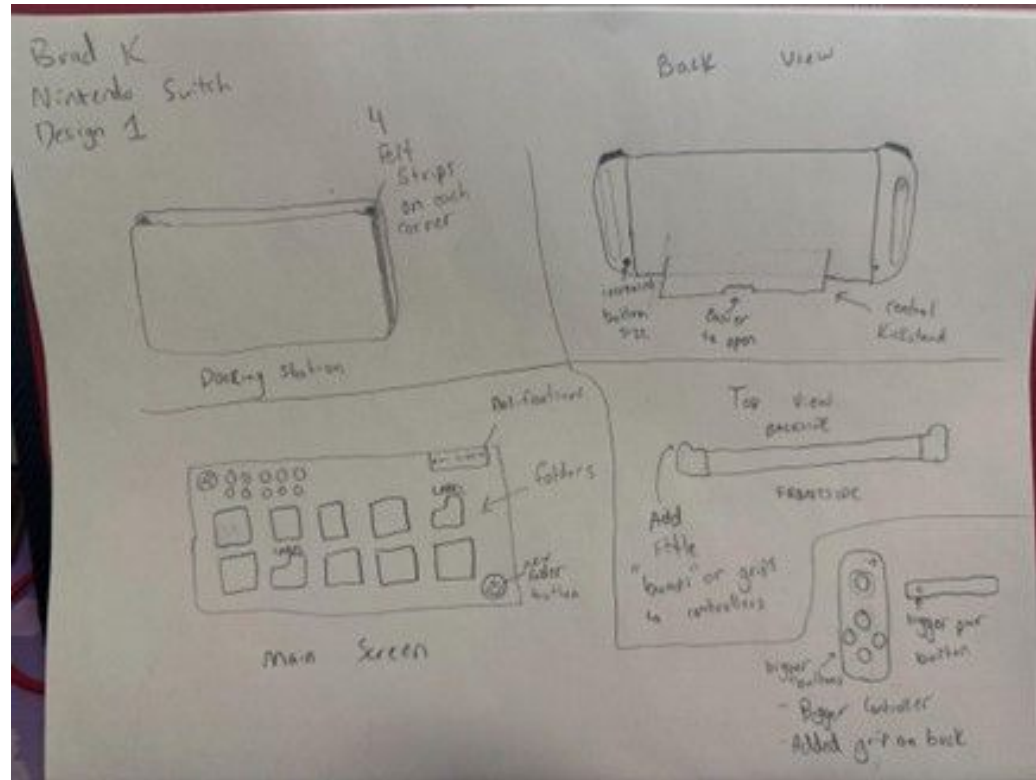
Sketches (Continued)



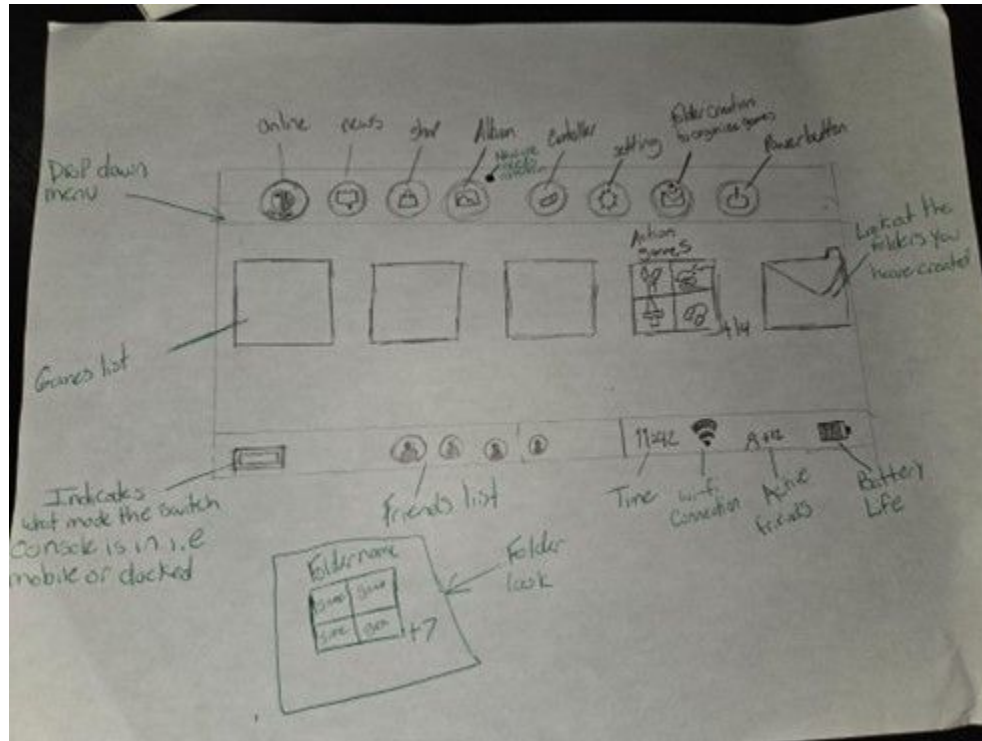
Sketches (Continued)



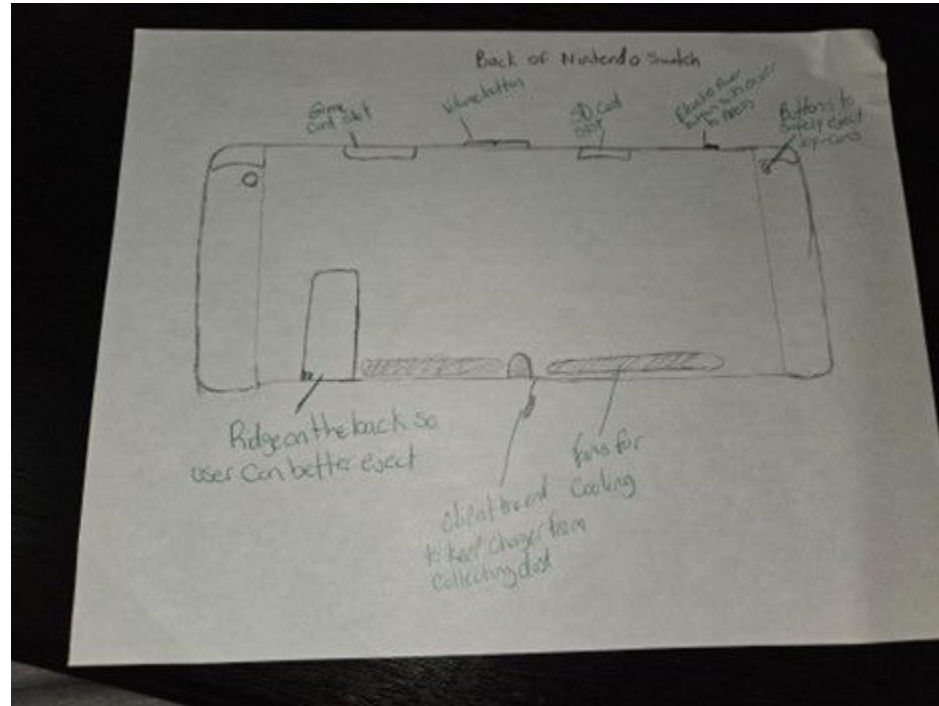
Sketches (Continued)



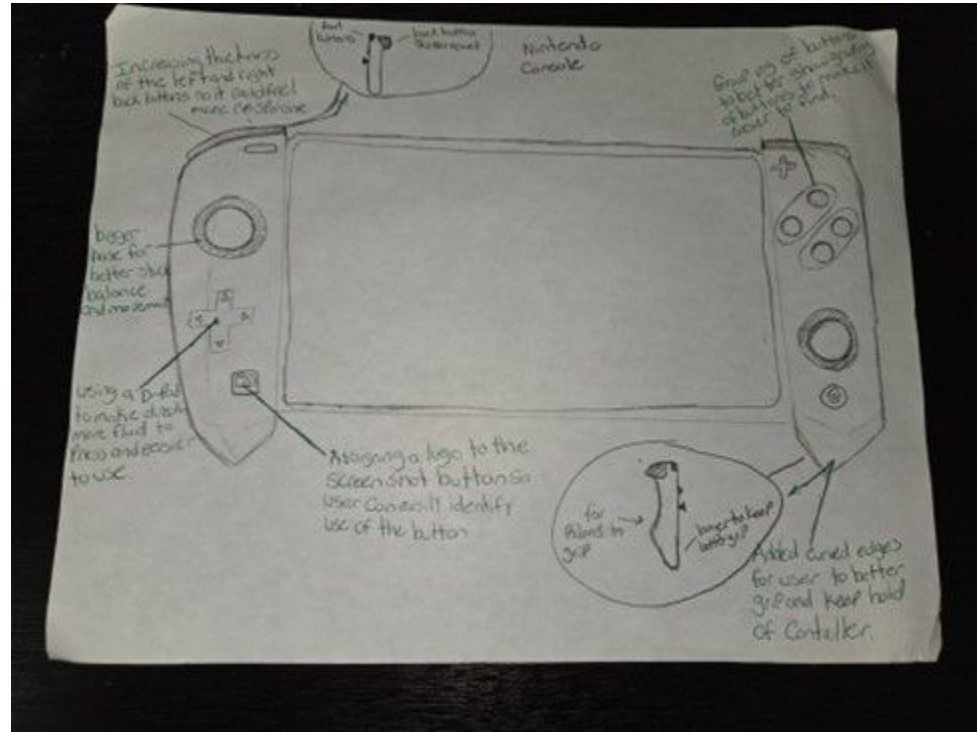
Sketches (Continued)



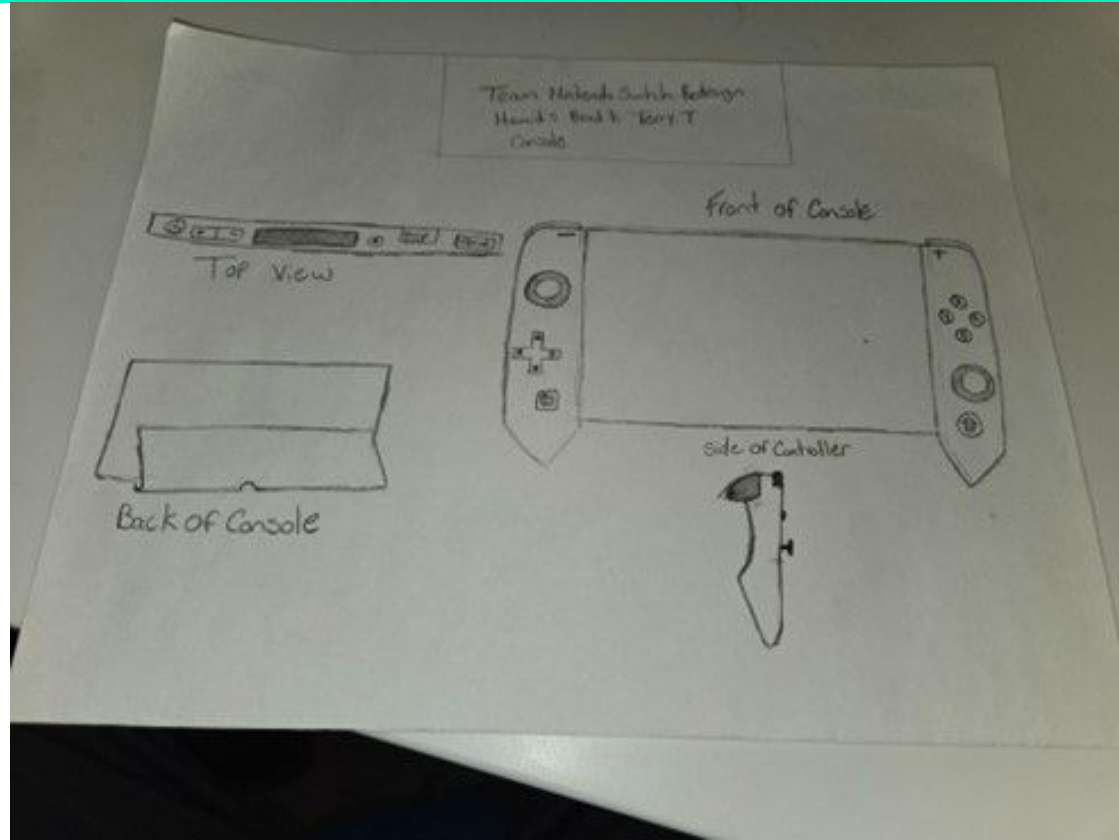
Sketches (Continued)



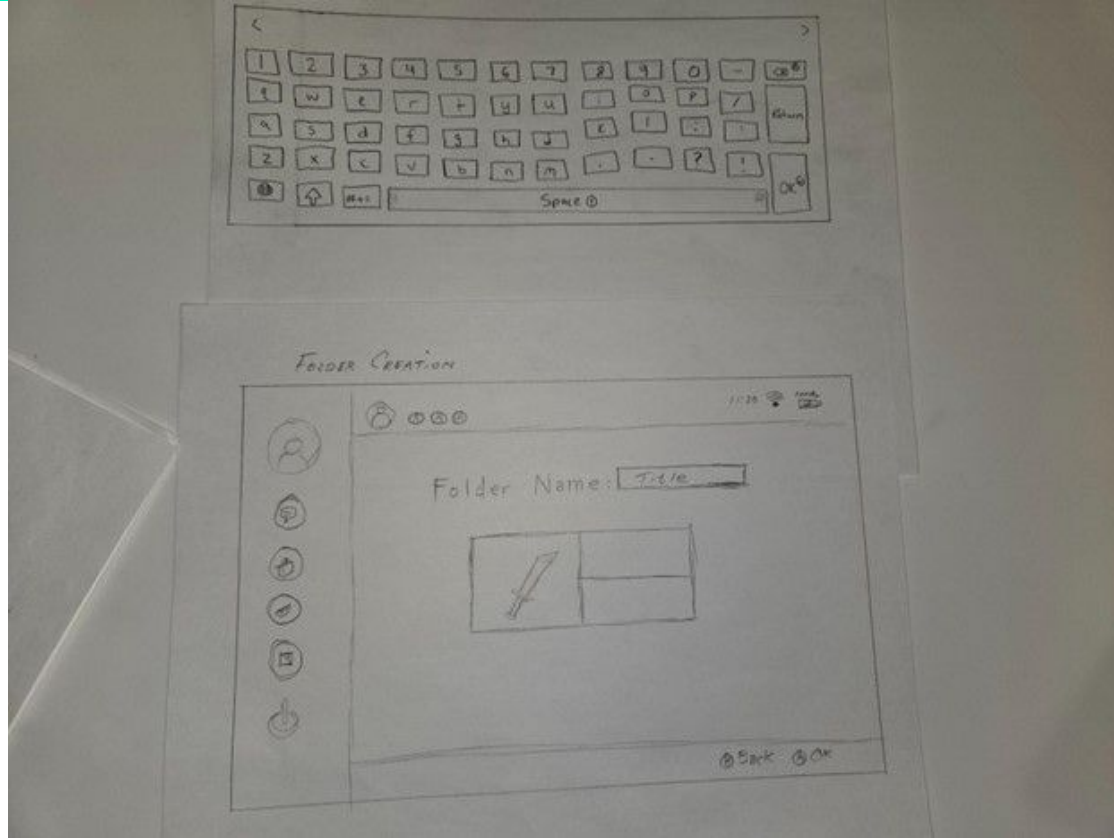
Sketches (Continued)



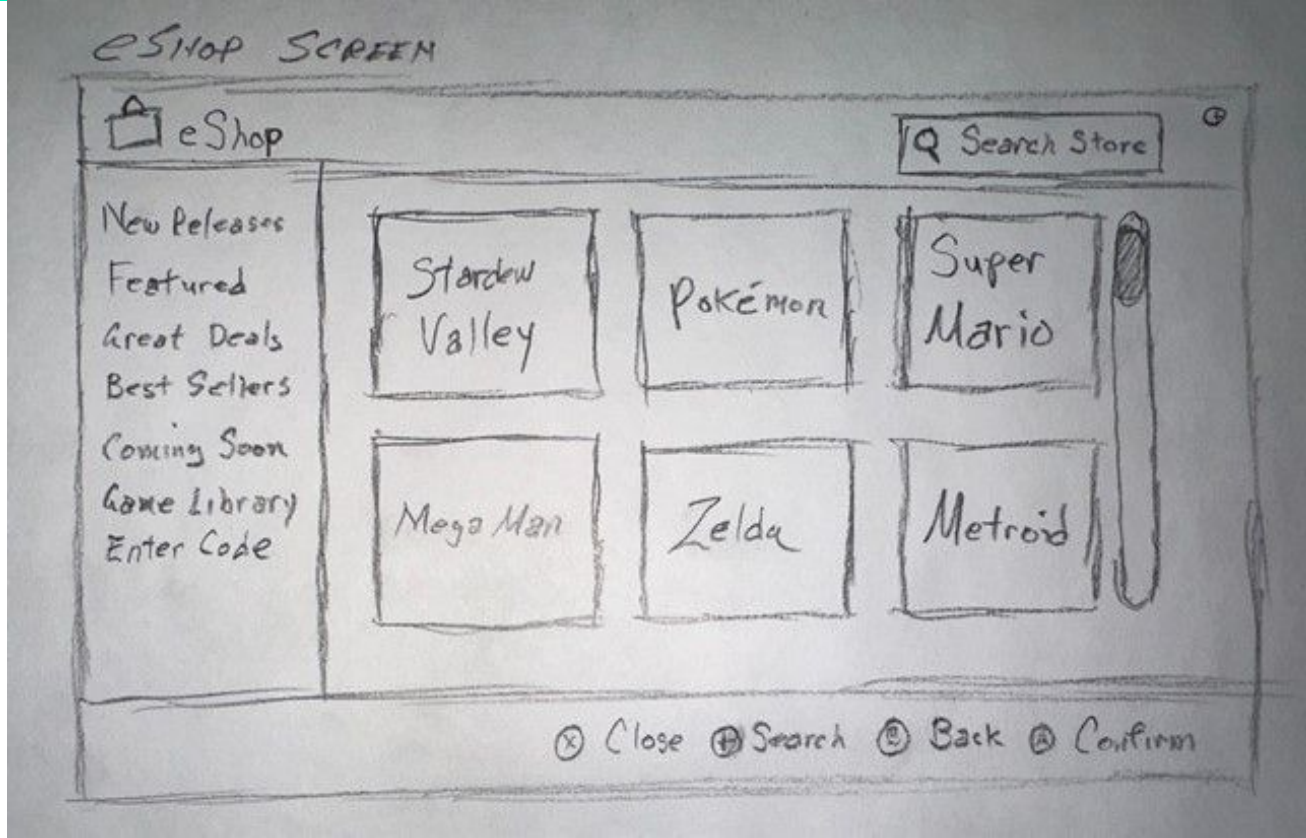
Final Paper Prototype (Hardware)



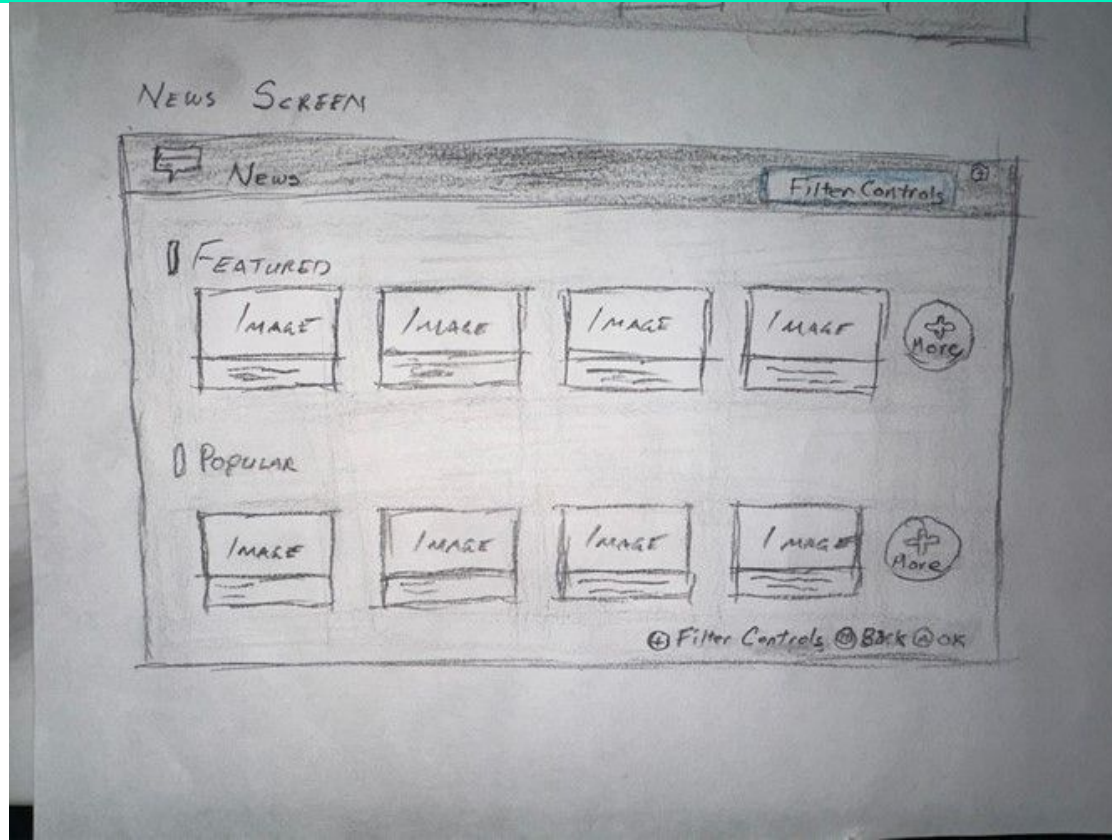
Final Paper Prototype (Software)



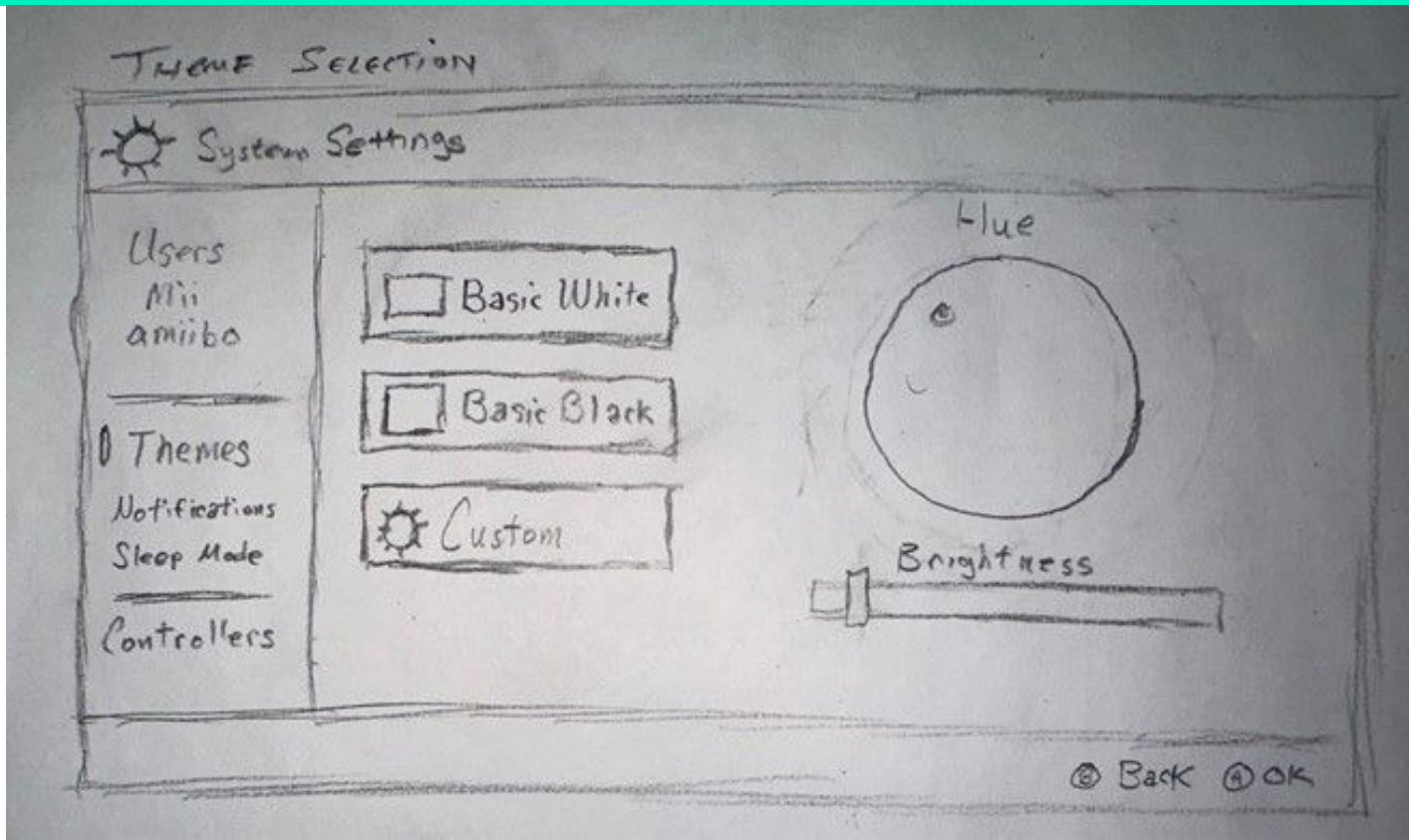
Final Paper Prototype (Software)



Final Paper Prototype (Software)



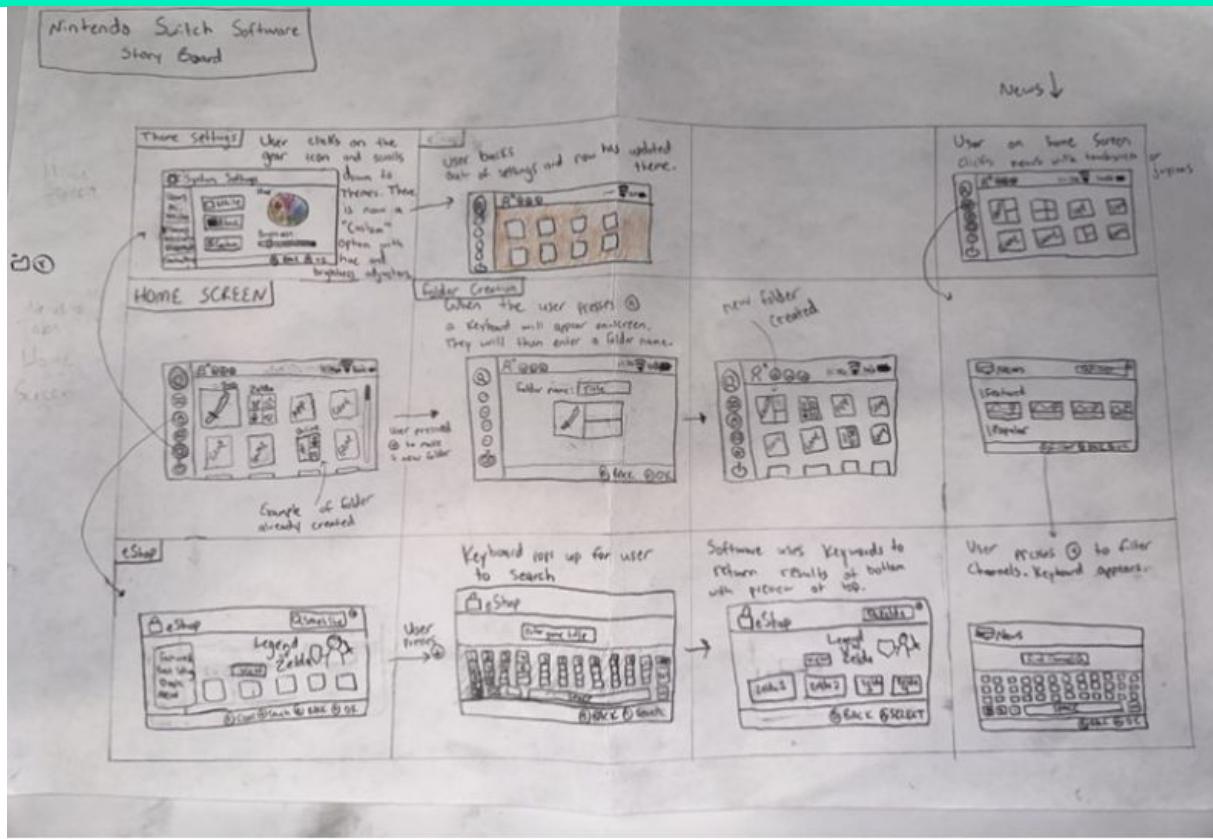
Final Paper Prototype (Software)



A decorative graphic consisting of two parallel, dark gray diagonal lines that extend from the top-left corner towards the bottom-left corner of the frame.

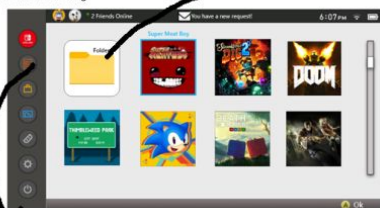
Storyboard

Storyboard

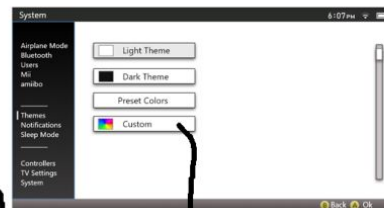


Storyboard

All buttons have been moved to the left-hand-side with the main page being listed in groups of 4 with vertical scrolling.



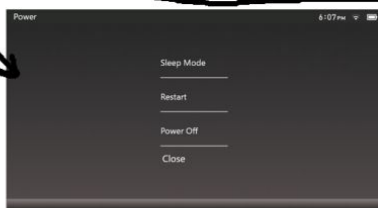
Game folders have been added for better organization of system's games.



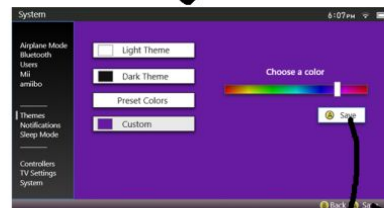
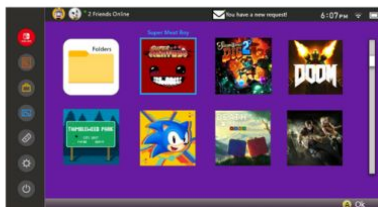
Find channels button is easily noticeable now with "Search" button added to button guide.



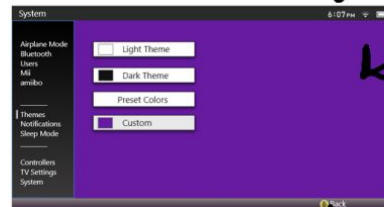
Search bar has been moved from left-side to the right for better visibility. We've also added the search button to the button guide on the bottom.



Extra options have been added to the power menu button. User can now put it in sleep mode, power off, restart, or close the power menu.



User can choose any custom color from the bar or preset colors in the button above.



After selecting a color and saving it, the user can return to the home screen to see changes.



Usability Evaluation

Usability Evaluation

- Icons should be bigger
- Difficult to find how to create a folder
- News button hard to identify
- Rename “Filter Controls” to “Find Channels”
- Difficulty using hue changer
- Remove brightness slider for themes

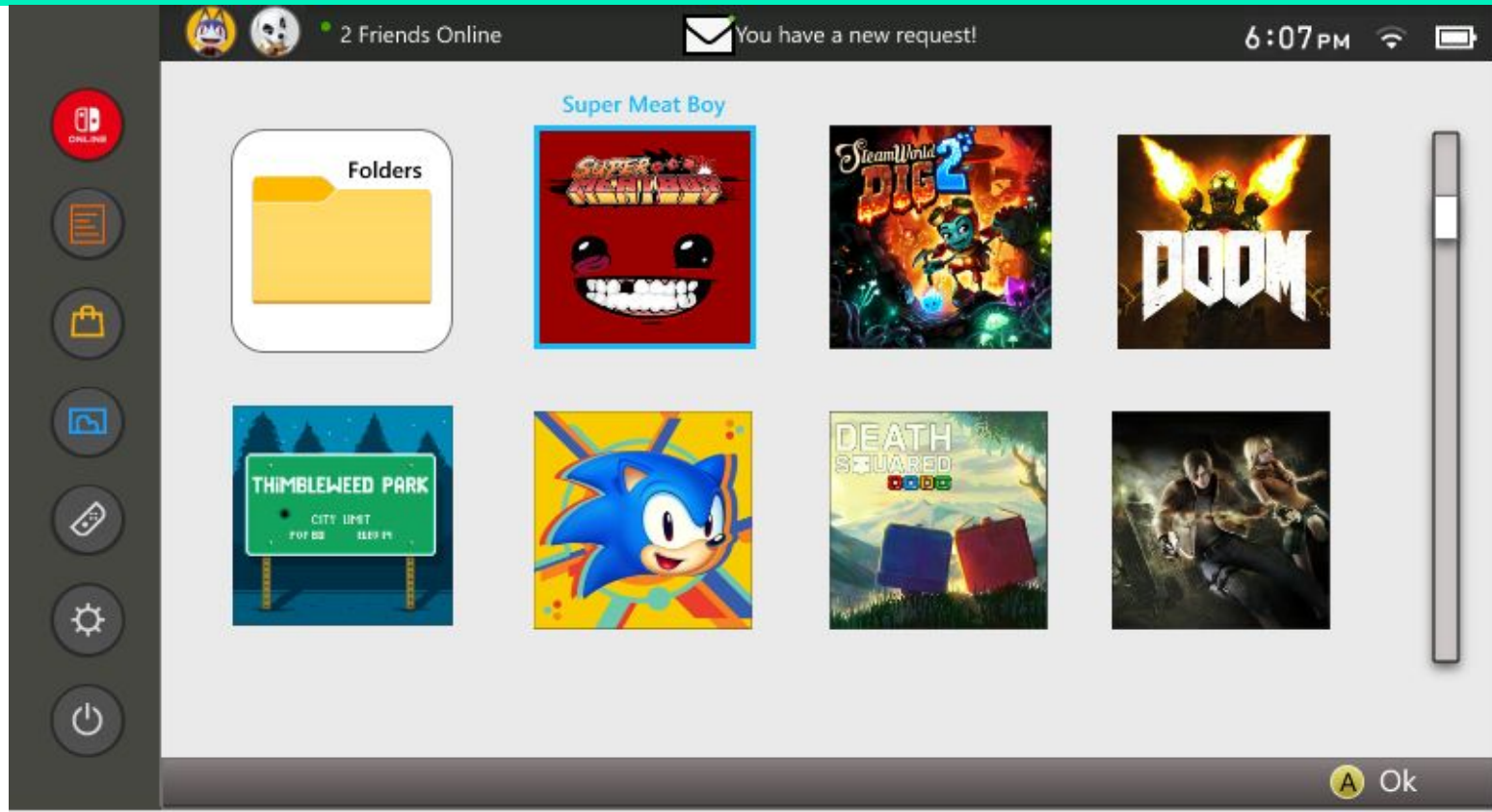
Changes from Usability Evaluation

- Bigger Icons
- Clearer method for creating folders
- Changed News Button Icon
 - Hover (not implemented)
- Renamed search bar on News Page
- Changed method of picking custom theme color
- Removed brightness slider for custom themes

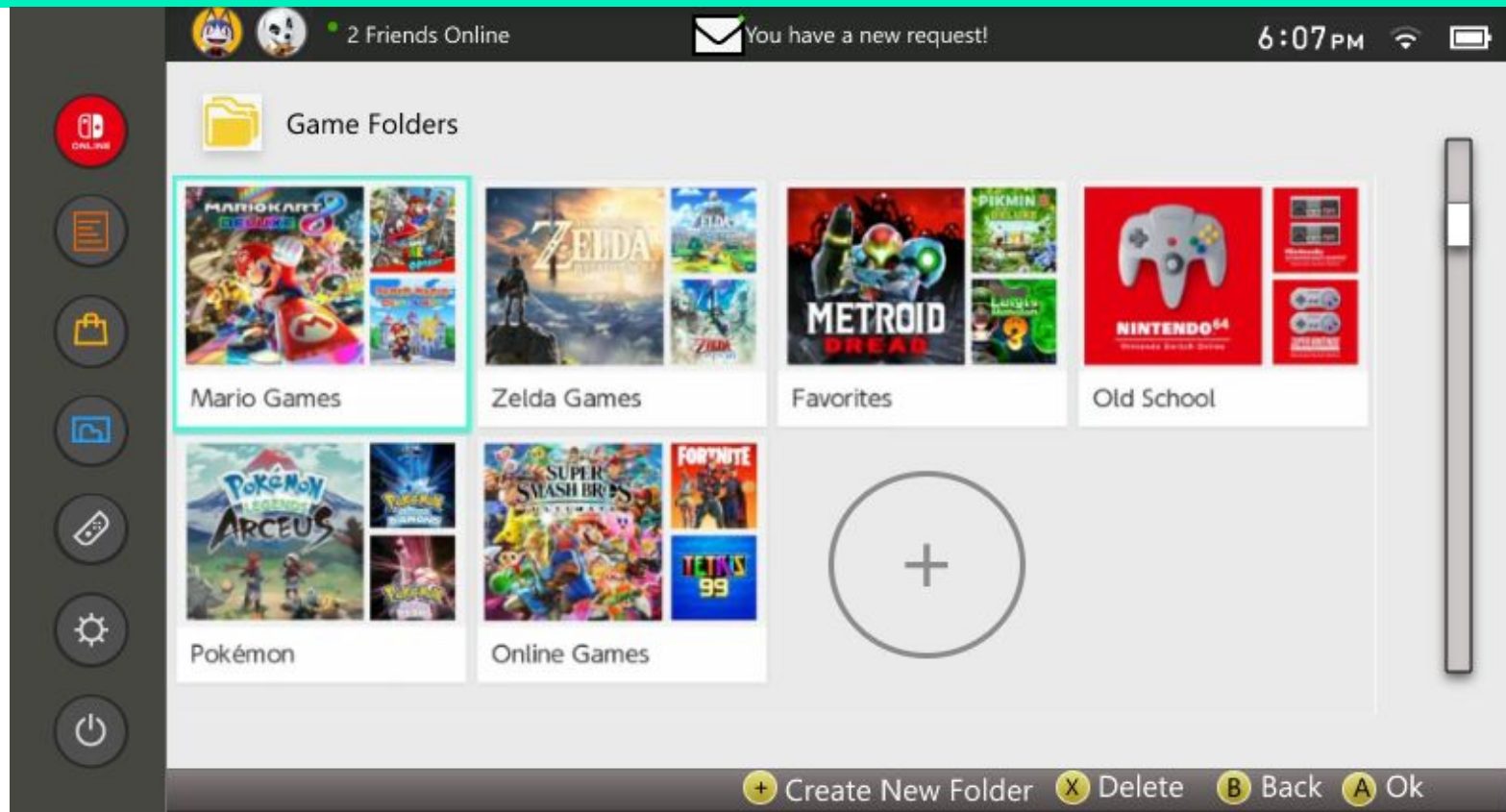


Working Prototype

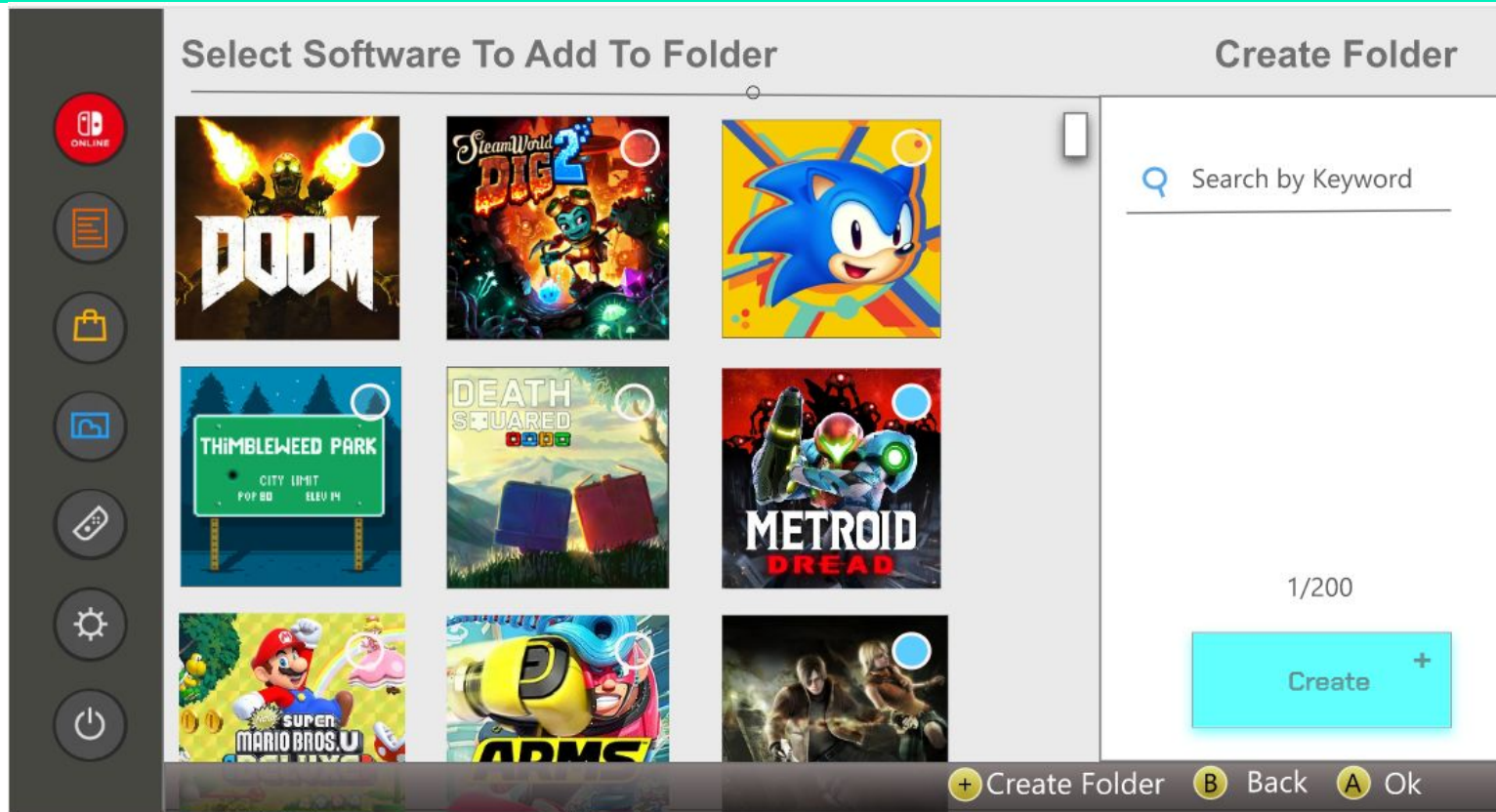
Working Prototype



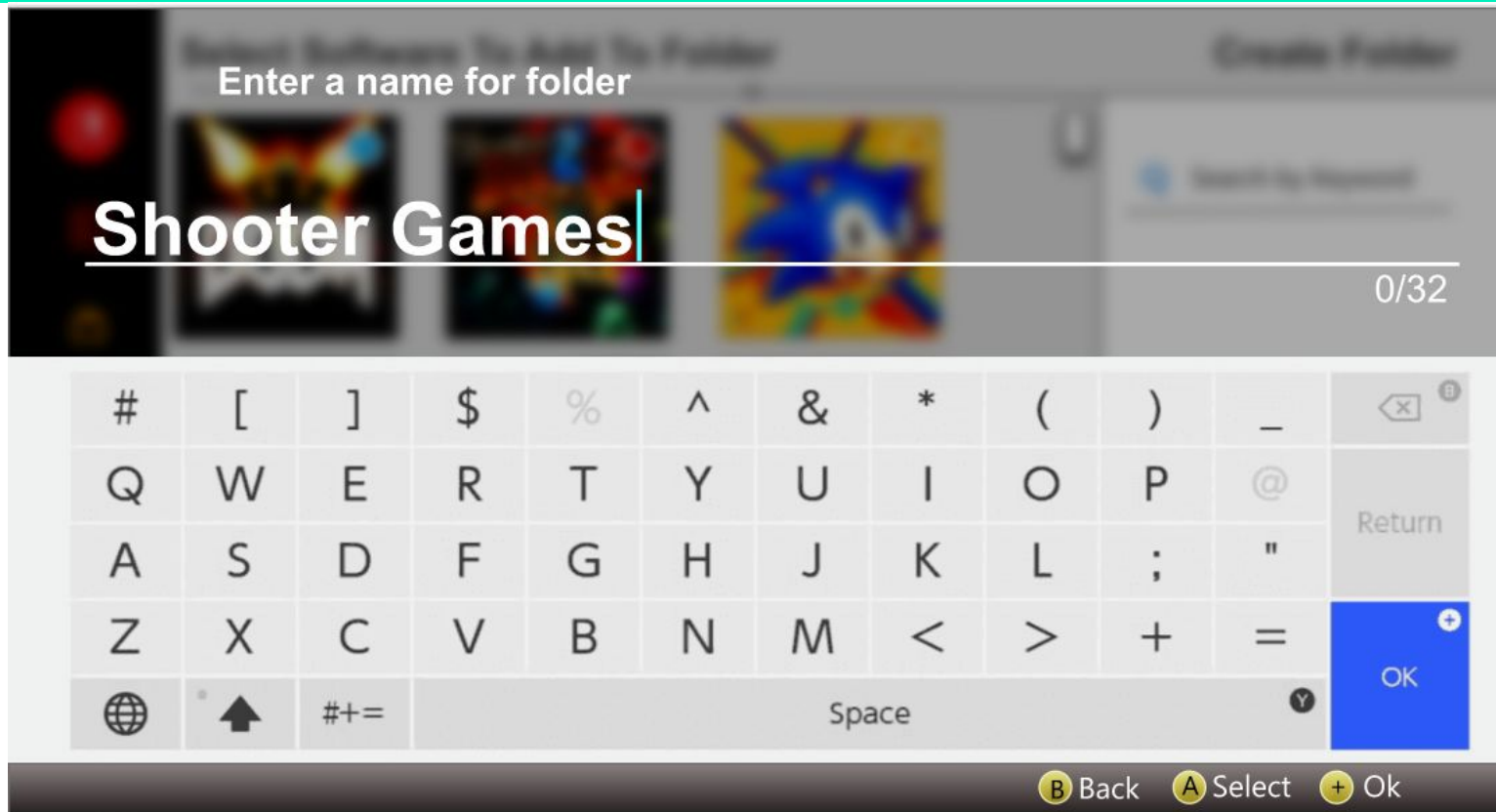
Working Prototype (Continued)



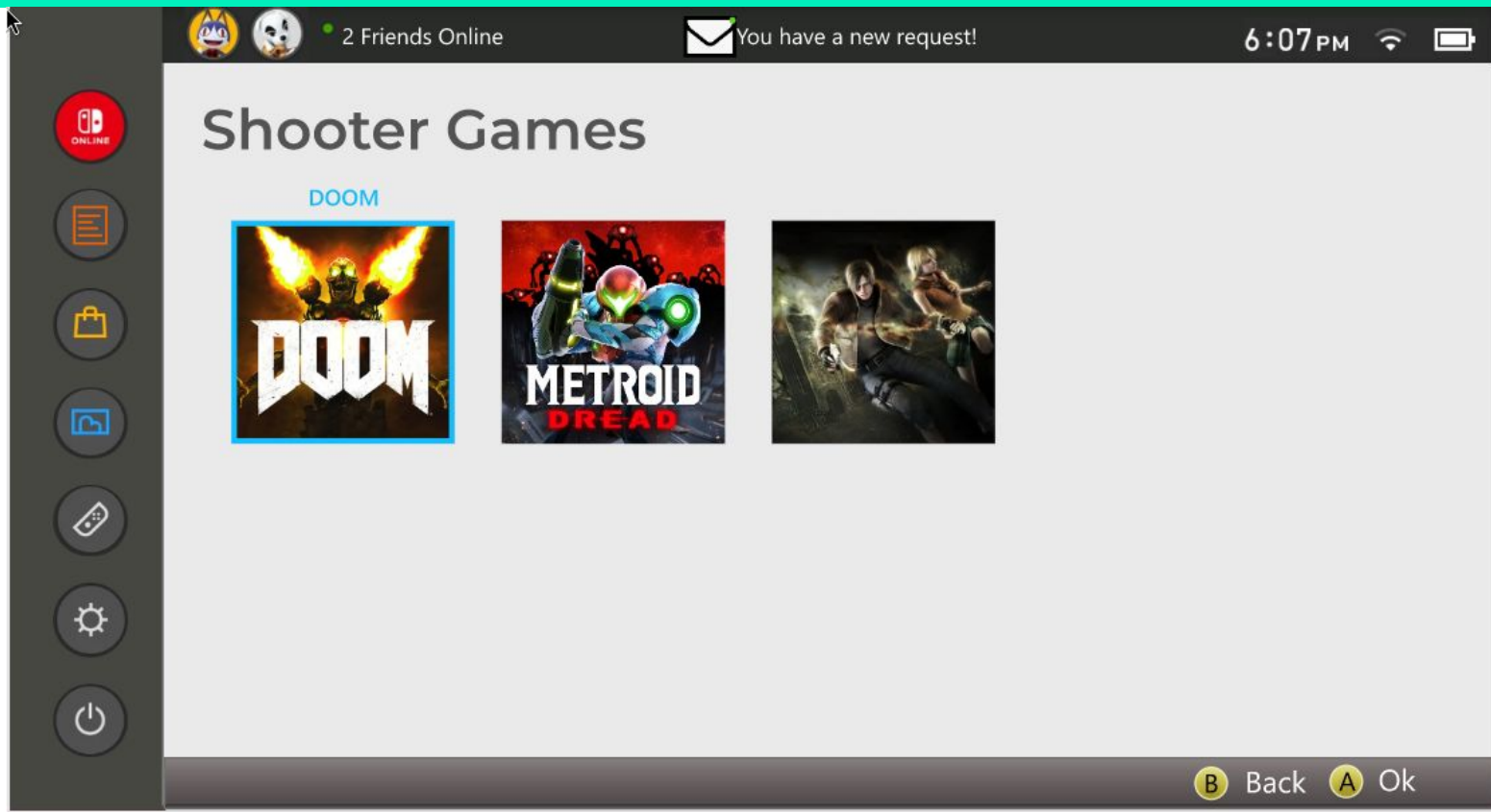
Working Prototype (Continued)



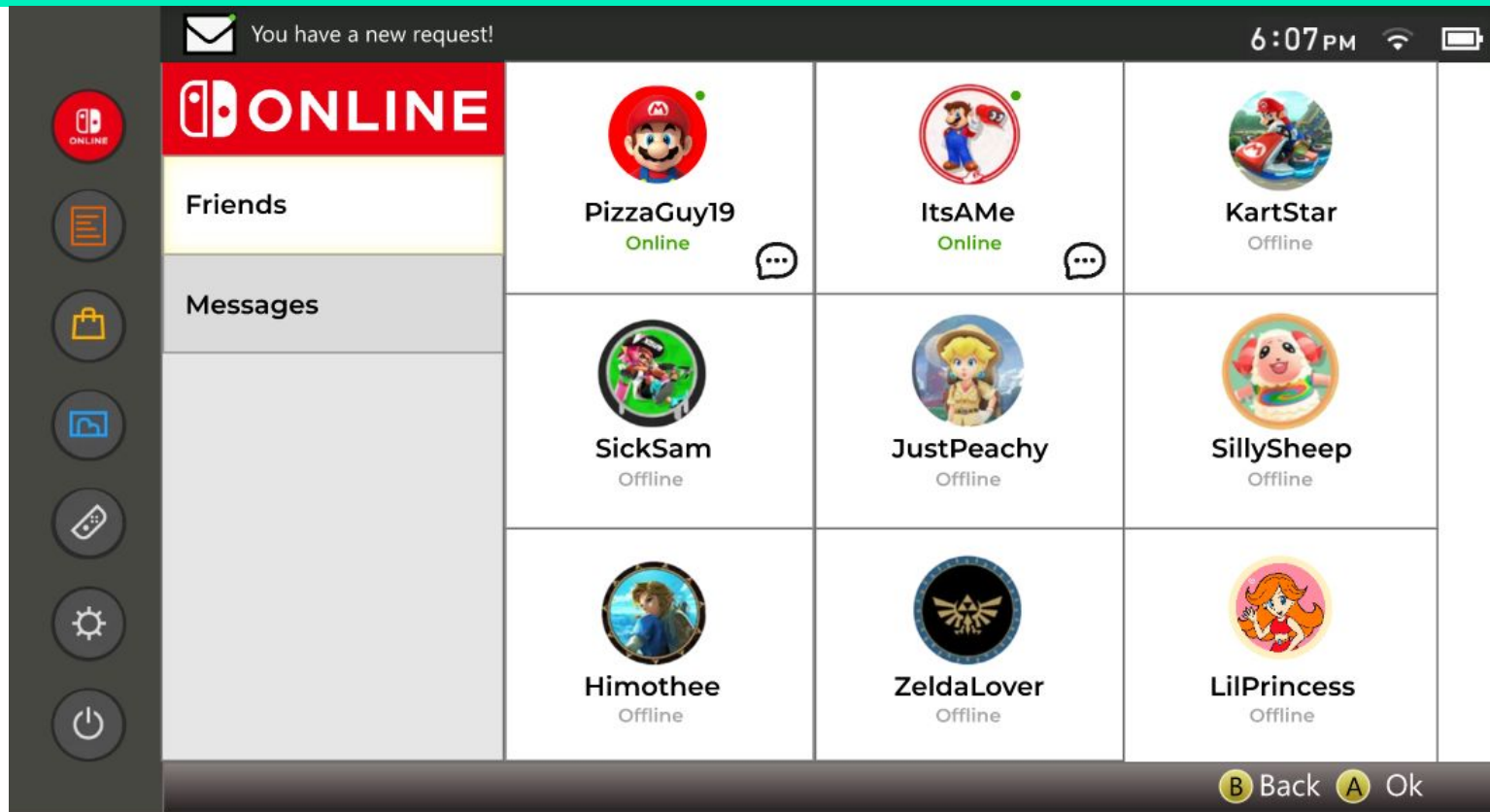
Working Prototype (Continued)



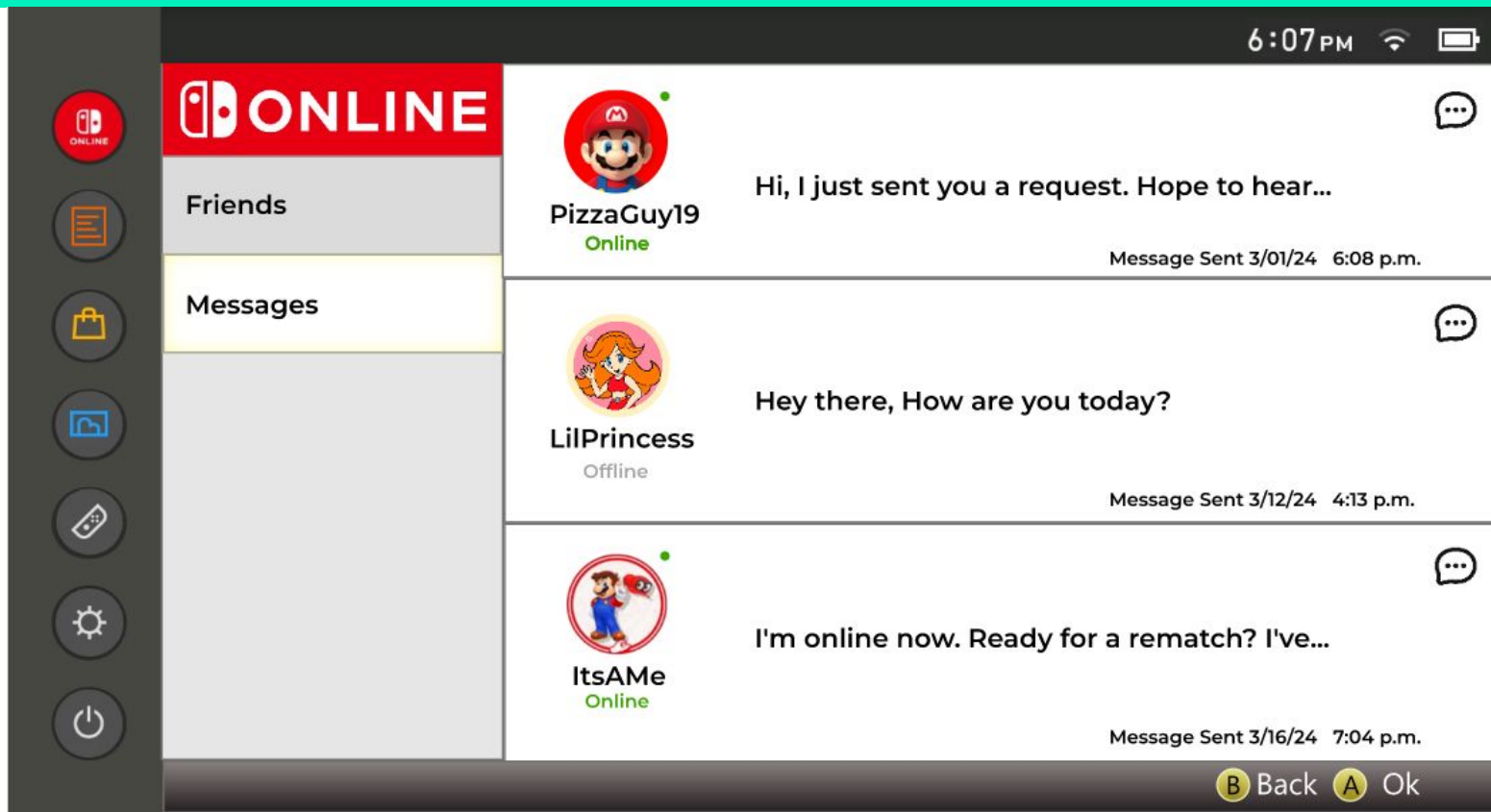
Working Prototype (Continued)



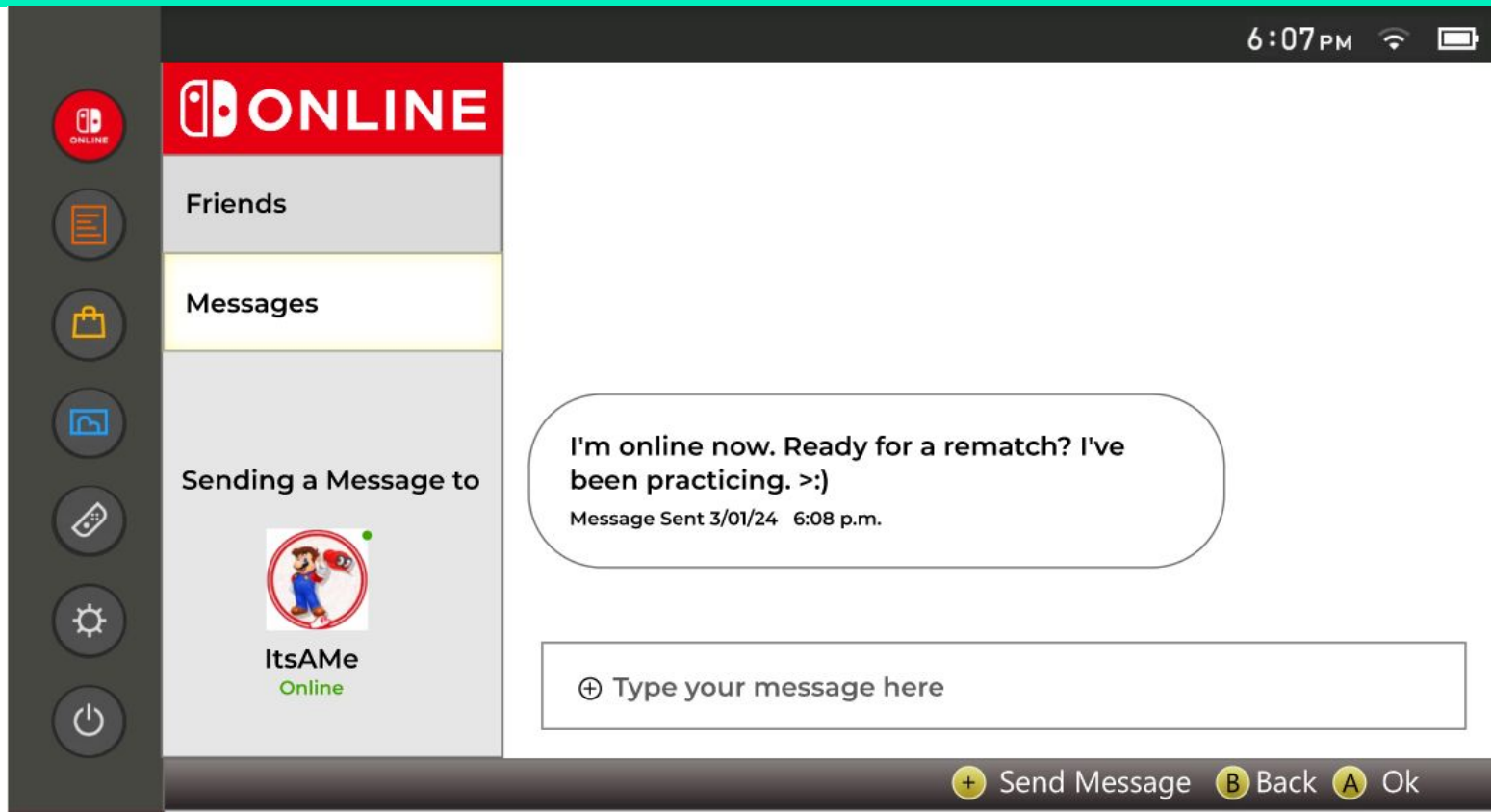
Working Prototype (Continued)



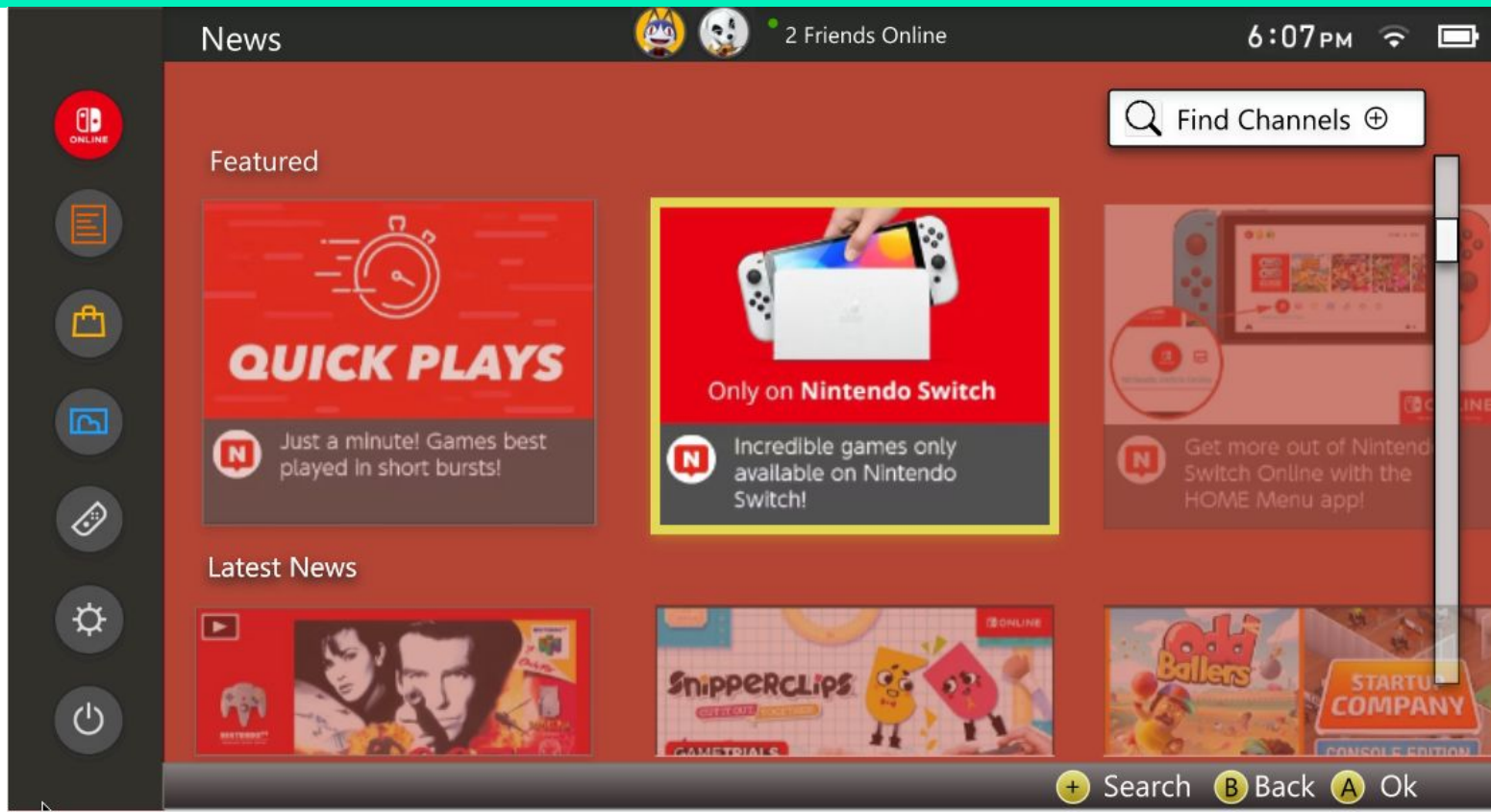
Working Prototype (Continued)



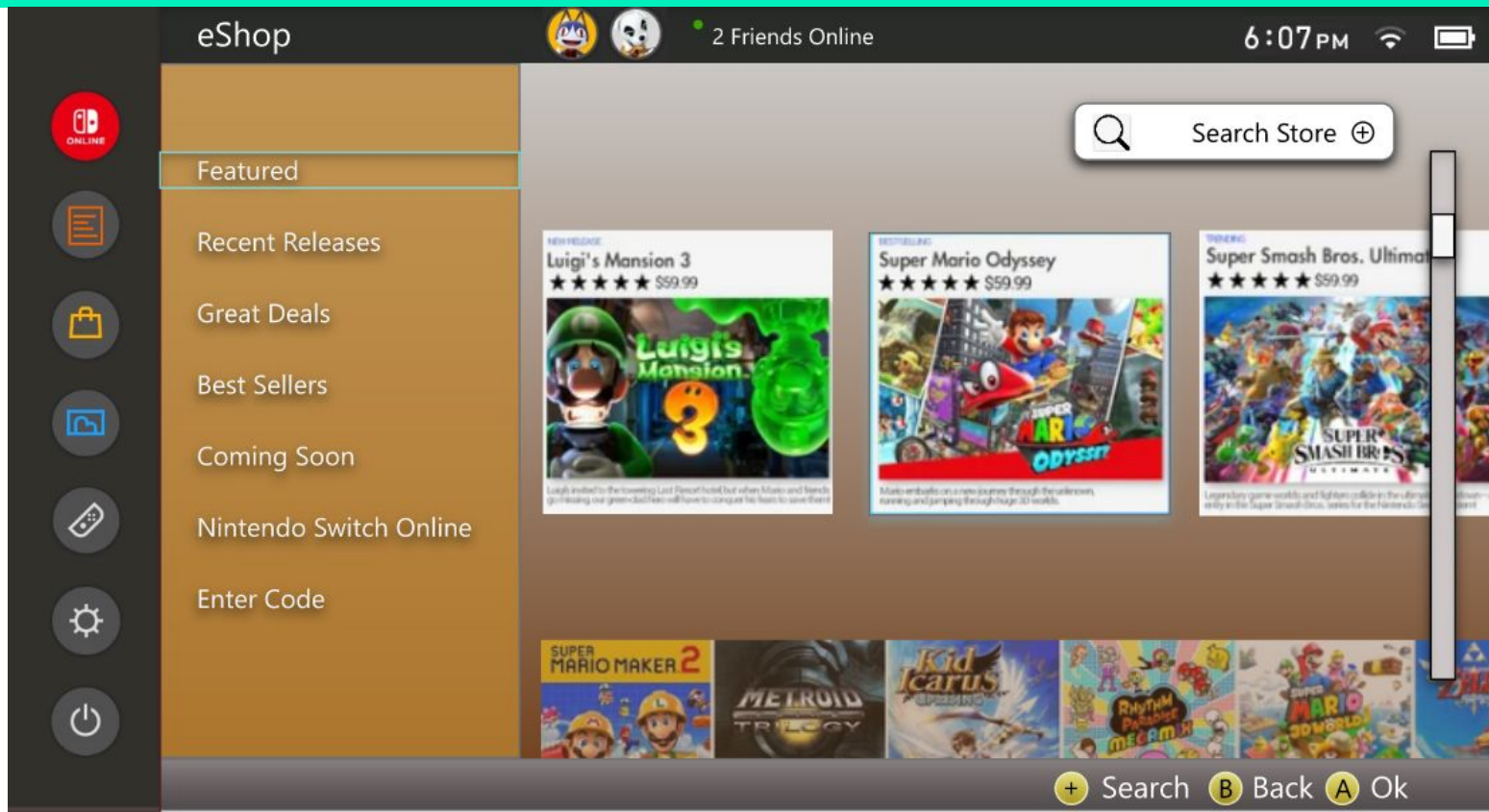
Working Prototype (Continued)



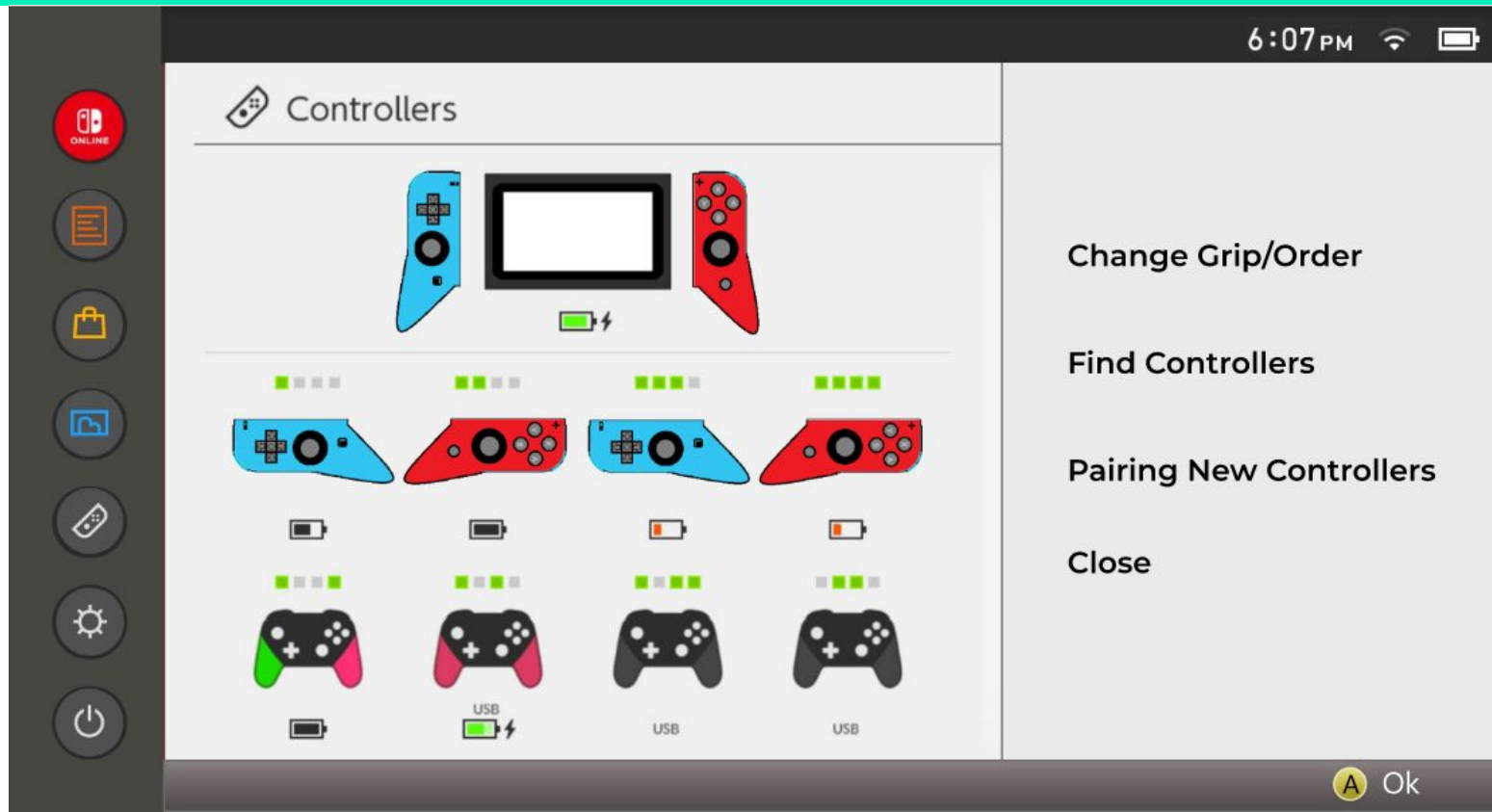
Working Prototype (Continued)



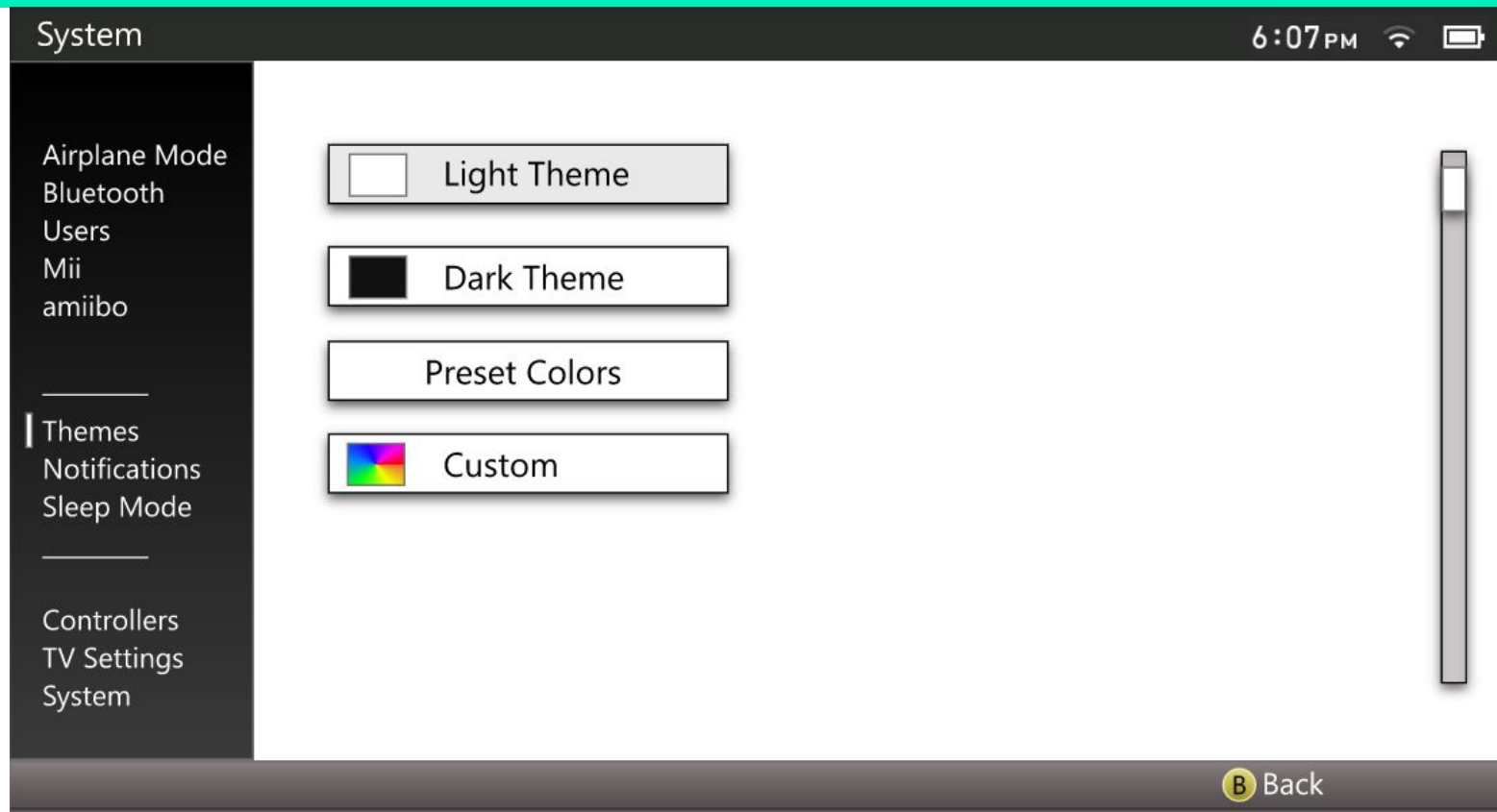
Working Prototype (Continued)



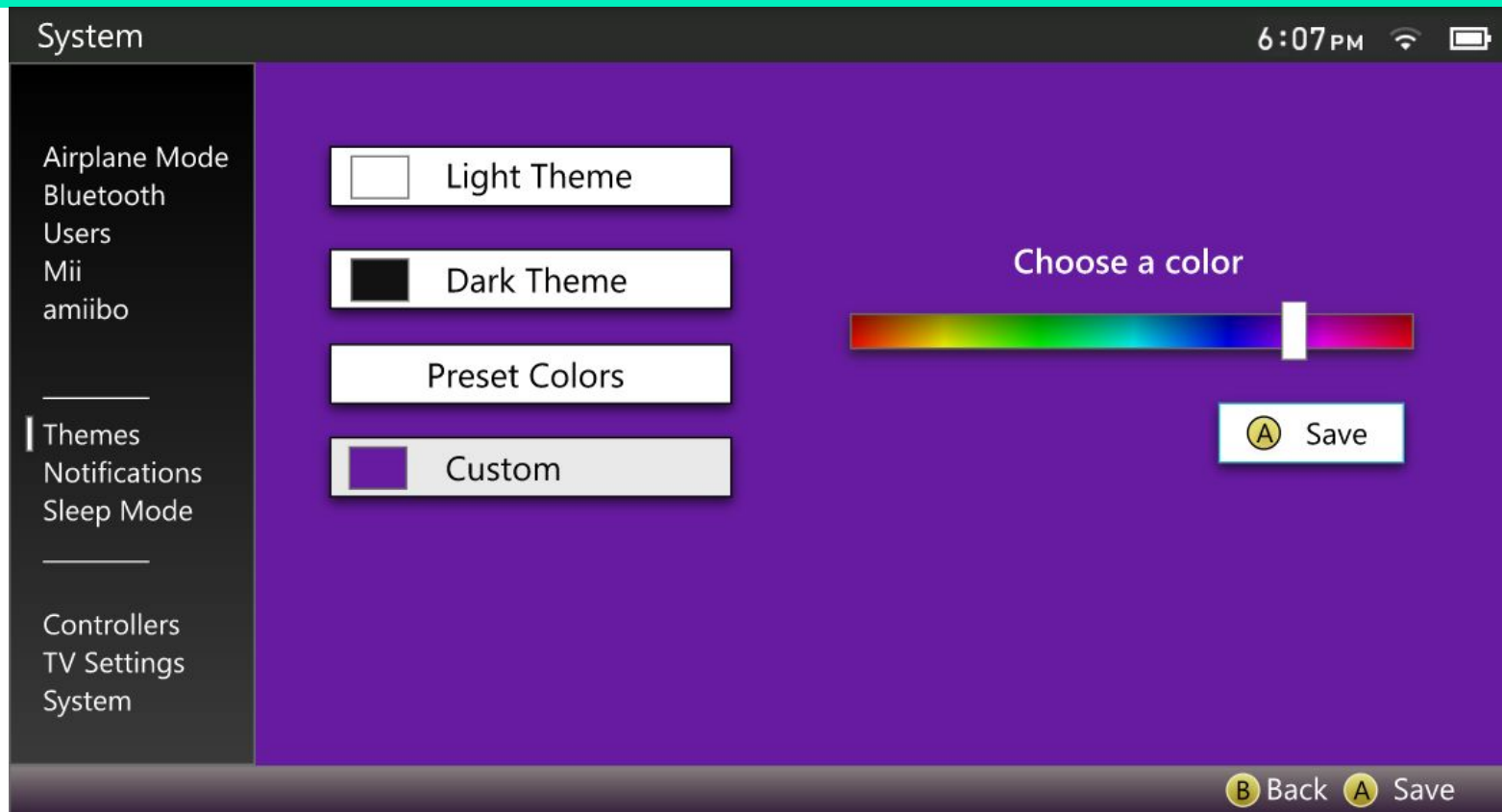
Working Prototype (Continued)



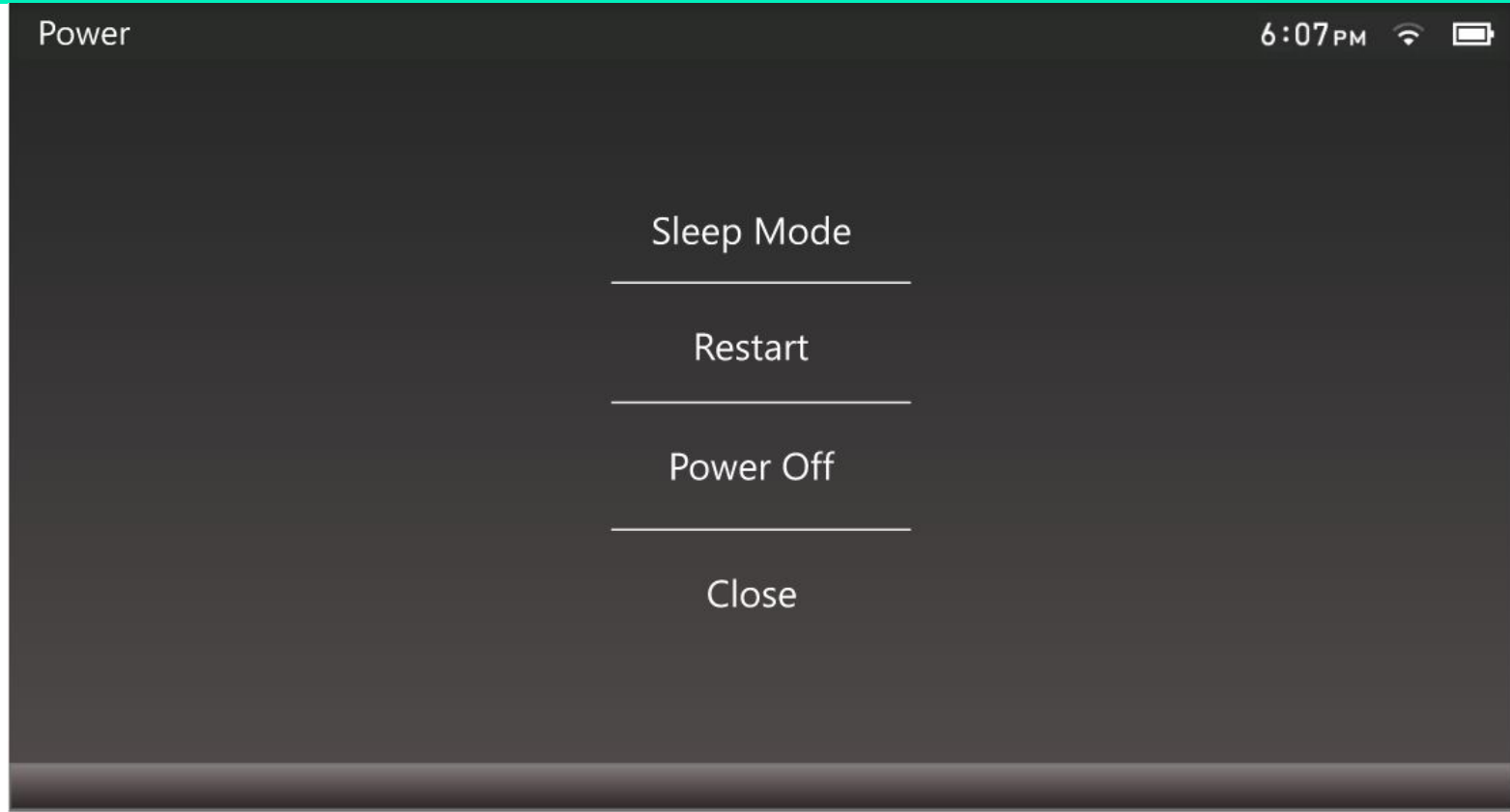
Working Prototype (Continued)



Working Prototype (Continued)



Working Prototype (Continued)





Heuristic Evaluation

Heuristic Evaluation

- Evaluated by Team LinkedIn
 - Home menu button to got to home screen at any point
 - Cluttered settings menu
 - Sleep mode setting in the system settings tab
 - Comment icon in messages tab not being necessary
 - Folders tab in the home screen being out of place
 - Two search options on the shop page being unnecessary
 - Formatting issues with the back button

Heuristic Evaluation (Continued)

- Two buttons to create a folder in game folders tab is unnecessary
- Corresponding lights for 8 Controllers is confusing to understand
- No indicator that user has changed the background color they have selected
- Plus button in the search store bar in the e-shop menu
- Consistency issues when trying to go back in the controller page
- No sidebar in the settings menu
- No button labels in power options screen
- No option to add friends or accept or deny friend requests

Modifications and Justifications

- We added a home screen option so user can use touch screen to go to home page at any time
- We made more space between the options in the settings menu
- We removed sleep mode option in settings menu
- We removed the icon in the messages tab on the friends screen
- We added a small icon in the corner of the home page to view game folders
- We only made one search store option in e-shop menu
- Tried our best to keep formatting with button labels consistent

Modifications and Justifications (Continued)

- Removed the create folder button in line with the other folders
- Limited the lights to just 4 controllers to make it easier to understand
- Kept in consideration a save changes option but could not be implemented due to time constraints
- Removed plus button in search store bar
- Replaced the 'ok' button with a 'back' to go back to previous page in controllers menu
- Decided to not add sidebar in settings menu to keep more space
- Added button labels in power options menu
- Made an add friends tab along with other tabs to fill space



Demo

<https://xd.adobe.com/view/e1be7a75-d411-4544-8764-9ffa597284ca-b4ed/>